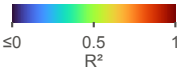
Pooled R², cruise held out

		Granite embedding	In-situ temperature	Time & place	Raw satellite	Main carrier, share of genes (one example metagenome)
Granite clearly better						28 shown of 360 KOs
Transport	**pstS** phosphate transport system substrate-binding…	0.46	0.25	0.11	0.45	Pseudomonadota 59%
	pstB phosphate transport system ATP-binding protein	0.46	0.17	0.12	0.43	Pseudomonadota 68%
	pstA phosphate transport system permease protein	0.42	0.16	0.15	0.42	Pseudomonadota 60%
	pstC phosphate transport system permease protein	0.39	0.08	0.20	0.36	Pseudomonadota 69%
	thiY putative hydroxymethylpyrimidine transport s…	0.38	0.06	-0.09	0.31	Pseudomonadota 84%
Amino acids	**pheA2** prephenate dehydratase	0.47	0.20	0.10	0.40	Pseudomonadota 43%
	glsA glutaminase	0.46	0.17	-0.22	0.48	Eukaryotes 35%
	E2.5.1.54 3-deoxy-7-phosphoheptulonate synthase	0.44	0.16	-0.13	0.40	Pseudomonadota 46%
	LACC1 purine-nucleoside/S-methyl-5'-thioadenosine…	0.44	0.23	0.03	0.39	Pseudomonadota 50%
	lysA diaminopimelate decarboxylase	0.43	0.06	-0.03	0.31	Pseudomonadota 49%
	argH argininosuccinate lyase	0.42	0.10	-0.00	0.34	Pseudomonadota 49%
	argG argininosuccinate synthase	0.40	0.06	0.01	0.34	Pseudomonadota 46%
	proA glutamate-5-semialdehyde dehydrogenase	0.40	0.01	0.09	0.31	Pseudomonadota 55%
Nucleotides	**PFAS** phosphoribosylformylglycinamidine synthase	0.46	0.22	-0.11	0.42	Pseudomonadota 44%
	purK 5-(carboxyamino)imidazole ribonucleotide syn…	0.42	0.16	0.02	0.34	Pseudomonadota 41%
Carbohydrate, lipid & cofactors	**glcF** glycolate dehydrogenase iron-sulfur subunit	0.45	0.23	-0.65	0.33	Pseudomonadota 57%
	E4.1.3.25 (S)-citramalyl-CoA lyase	0.44	0.16	-0.05	0.56	Pseudomonadota 83%
	nadA quinolinate synthase	0.40	0.12	0.02	0.34	Pseudomonadota 41%
	dhaA haloalkane dehalogenase	0.40	0.25	-0.02	0.37	Pseudomonadota 63%
	ispH 4-hydroxy-3-methylbut-2-en-1-yl diphosphate…	0.40	0.15	-0.15	0.34	Pseudomonadota 49%
	ENO1_2_3 enolase 1/2/3	0.40	0.20	-0.19	0.32	Pseudomonadota 46%
	pntA proton-translocating NAD(P)+ transhydrogenas…	0.39	0.24	-0.18	0.35	Pseudomonadota 51%
DNA, RNA & protein synthesis	**scpA** segregation and condensation protein A	0.48	0.27	0.00	0.41	Pseudomonadota 49%
	FARSA phenylalanyl-tRNA synthetase alpha chain	0.41	0.18	-0.16	0.34	Pseudomonadota 42%
	udg uracil-DNA glycosylase	0.40	0.15	-0.01	0.36	Pseudomonadota 53%
	CARS cysteinyl-tRNA synthetase	0.39	0.16	-0.31	0.35	Pseudomonadota 43%
Other / uncharacterised	**tldD** TldD protein	0.43	0.25	-0.23	0.43	Pseudomonadota 39%
	K06893 uncharacterised protein	0.42	0.13	0.05	0.32	Pseudomonadota 70%
Both high, about equal						14 shown of 37 KOs
Energy & photosynthesis	**hoxH** NAD-reducing hydrogenase large subunit	0.71	0.75	-0.31	0.53	Cyanobacteria 99%
	ndhD NAD(P)H-quinone oxidoreductase subunit 4	0.71	0.76	-0.27	0.49	Cyanobacteria 98%
	ndhK NAD(P)H-quinone oxidoreductase subunit K	0.70	0.70	-0.26	0.45	Cyanobacteria 100%
	ndhJ NAD(P)H-quinone oxidoreductase subunit J	0.69	0.73	-0.27	0.44	Cyanobacteria 90%
	cpeS phycoerythrin-associated linker protein	0.67	0.68	-0.22	0.49	Cyanobacteria 99%
Transport	**mntB** manganese transport system permease protein	0.70	0.69	-0.13	0.54	Cyanobacteria 66%
	mntC manganese transport system substrate-binding…	0.69	0.66	-0.16	0.44	Cyanobacteria 100%
	ecfT energy-coupling factor transport system perm…	0.67	0.67	-0.12	0.47	Cyanobacteria 41%
Nucleotides	**nrdJ** ribonucleotide reductase, class II	0.67	0.68	-0.23	0.45	Cyanobacteria 100%
Carbohydrate, lipid & cofactors	**K14330** fatty aldehyde-generating acyl-ACP reductase	0.69	0.71	-0.21	0.50	Cyanobacteria 100%
	chlL light-independent protochlorophyllide reduct…	0.65	0.69	-0.15	0.48	Pseudomonadota 65%
	E3.1.2.28 1,4-dihydroxy-2-naphthoyl-CoA hydrolase	0.65	0.67	-0.31	0.42	Cyanobacteria 78%
Other / uncharacterised	**K07015** Rab family protein	0.67	0.70	-0.27	0.45	Cyanobacteria 100%
	mviM virulence factor MviM	0.67	0.68	-0.15	0.46	Cyanobacteria 59%
Temperature clearly better						12 shown of 280 KOs
Energy & photosynthesis	**ndhF** NAD(P)H-quinone oxidoreductase subunit 5	0.53	0.65	-0.24	0.33	Cyanobacteria 95%
	puhA photosynthetic reaction center H subunit	0.31	0.44	-0.32	0.42	Pseudomonadota 74%
	pufL photosynthetic reaction center L subunit	0.31	0.44	-0.35	0.44	Pseudomonadota 83%
Transport	**fiu** catecholate siderophore receptor	0.32	0.51	-0.45	0.35	Verrucomicrobiota 58%
Nucleotides	**RRM2** ribonucleoside-diphosphate reductase subunit…	0.27	0.40	-0.60	0.38	dsDNA viruses 44%
Carbohydrate, lipid & cofactors	**dgkA** diacylglycerol kinase (ATP)	0.45	0.57	-0.94	0.44	Eukaryotes 44%
	GPAT3_4 glycerol-3-phosphate O-acyltransferase 3/4	0.34	0.45	-0.84	0.44	Eukaryotes 99%
	ams amylosucrase	0.35	0.45	-0.40	0.40	Cyanobacteria 88%
DNA, RNA & protein synthesis	**rex** redox-sensing transcriptional repressor	0.40	0.56	-0.47	0.46	Planctomycetota 34%
	POLD1 DNA polymerase delta subunit 1	0.36	0.47	-0.66	0.47	dsDNA viruses 50%
	TOP2 DNA topoisomerase II	0.23	0.44	-0.65	0.41	dsDNA viruses 41%
	NOP2 25S rRNA (cytosine2870-C5)-methyltransferase	0.29	0.39	-0.79	0.35	Eukaryotes 96%



Best model so far: ridge regression on 32 principal components of the OSTIA-filled Granite embedding + in-situ temperature (median pooled R² 0.34 on the 20-KO benchmark and 0.12 over all 3,921 KOs, cruise held out; RESULTS.md R12, R17).
The heatmap compares the embedding alone (ridge), temperature alone (spline), time & place (latitude, longitude, day of year; ridge) and the raw satellite values (16 OLCI band means + SST mean of the tile, rank-normalised; spline).